Mathematical Induction → proves that a formula is true

$P(n)$

To prove that $P(n)$ is true for all positive integers n we need 2 steps

1) Basis step verify that $P(1)$ = true

2) Inductive step : show that conditional statements $P(k) \rightarrow P(k+1)$ is true for all positive integers k.

Expressed as a rule of inference:

$$[P(1) \wedge \forall k\, P(k) \rightarrow P(k+1)] \rightarrow \forall n\, P(n)$$

which $n \in Z^+$

$P(1)$ is true and if $P(k)$ is true (for every k) then $P(k+1)$ is true so it implies for every n $P(n)$ is true

e.g Prove that

$$1+2+3 \cdots + n = \frac{n(n+1)}{2}$$

Basis For $n=1$ $\quad \frac{1(1+1)}{2} = \boxed{1}$ ✓ $\quad 1=1$

Inductive

$n=k$ $\quad 1+2+3 \cdots + k = \frac{k(k+1)}{2}$

Adding $(k+1)$ $\quad 1+2+ \cdots + k+(k+1) = \frac{k(k+1)+(k+1)}{2}$

$$= \frac{k(k+1) + 2(k+1)}{2}$$

$$= \frac{(k+1)(k+2)}{2} \quad \text{factor out } k+1$$

$$= \frac{(k+1)((k+1)+1)}{2}$$

$n = k+1$ verified

e.g

$2^n < n!$ $\forall n$ $n \geq 4$

Basis

$n = 4$ $2^4 < 4!$ $16 < 24$ (True)

Inductive Assume $2^k < k!$ (multiply both sides by 2)

now proving for $2^{k+1} < (k+1)!$ $2^k \cdot 2 < 2 \cdot k!$ to bring $k+1$

$2^{k+1} < (k+1)!$

$3! = 3 \times 2 \times 1 = 6$

$4! = 4 \times 3 \times 2 \times 1 = 24$

$(k+1)!$ → multiply all numbers from 1 to $k+1$

$(k+1)! = (k+1) \times k!$

$k = 4$

$(4+1)! = 5! = 5 \times \underbrace{4 \times 3 \times 2 \times 1}_{4!}$

$5 \times 4!$

e.g $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} \cdots \frac{1}{2^n} = \frac{2^n \cdot 1}{2^n}$

(a) For $n=1$ $\frac{1}{2^1} = \frac{1}{2}$

For $n=2$ $\frac{1}{2^2} = \frac{1}{4}$

For $n=1$ $\frac{1}{2} = \frac{2^1 - 1}{2^1} = \frac{1}{2}$ ✓

$\frac{2^k - 1}{2^k}$ Adding $\frac{1}{2^{k+1}}$ on both sides

$$\frac{2^k - 1}{2^k} + \frac{1}{2^{k+1}} = \frac{(2^k - 1)2^{k+1}}{2^{k+1}}$$

$$= \frac{2^{2k+1} - 2^{k+1}}{2^{k+1}} \qquad \frac{2^{k+1}(2^k - 1)}{2^{k+1}}$$

133 divides $11^{n+1} + 12^{2n-1}$

Basis $n=1$ $\quad 11^{1+1} + 12^{2(1)-1} = 11^2 + 12^1 = 133$

true

$11^{k+1} + 12^{2k-1}$

$11^{(k+1)+1} + 12^{2(k+1)-1}$

$11^{(k+1)+1} + 12^{2k+2-1}$

$11^{k+1} \cdot 11 + 12^2 \cdot 12^{2k-1}$

$11^{k+1} \cdot 11 + 144 \cdot 12^{2k-1}$

$11^{k+1} \cdot 11 + (133+11)\,12^{2k-1}$

$11^{k+1} \cdot 11 + 12^{2k-1}(133) + 12^{2k-1}(11)$

$11\,(11^{k+1} + 12^{2k-1}) + 133\,(12^{2k-1})$

Prove according to this

Strong mathematicae induction:

The only difference is in induction process.

We assume that all of $P(1), P(2), \ldots P(k)$ are true to prove $P(k+1)$

Ch:- 6 counting:-

And then

Product Rule = $n_1 \times n_2$

(gives all possible combinations of task done in order.

3 Shirts (A, B, C)

2 pants (x, y)

$Ax + Ay + Bx + By + Cx + Cy$

S P 12

$S \times 12 + P \times 11$

(0 is neither pos/neg)

If you can describe the process as first do this then do that and both parts have their own number of options → multiply them using the product Rule

Sum Rule:- → use when you have a choice b/w two tasks and you can do only one of them

OR/either

$n_1 + n_2$

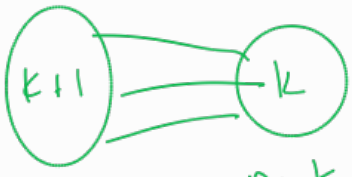
Subtraction Rule:-

$(n_1 + n_2) - (n_1 \cap n_2)$ → common

↓

you can do a task in two diff ways

pigeon hole :- We have more pigeons than pigeonholes
So atleast one hole must have two pigeons inside it.



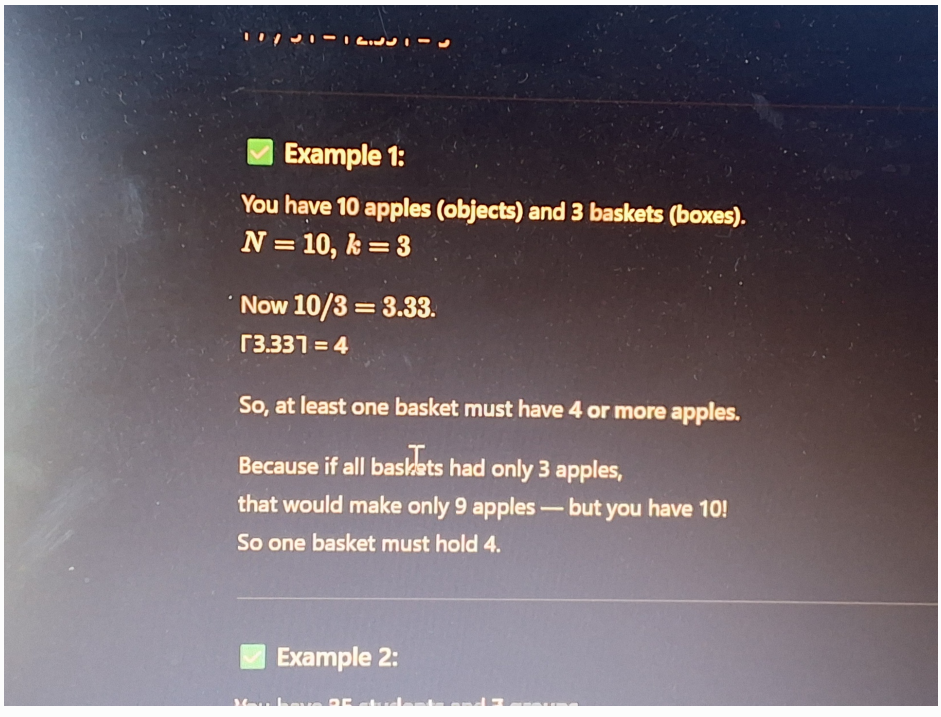
not one-one

generalised pigeon hole principle:-

$\lceil 7 \rceil = 7$

$\lceil 7.01 \rceil = 8$





Ex 6.2 Q4

bowl → 10 Red Balls
→ 10 blue balls

Total boxes = 20

(a) object → Balls
Boxes → Red / blue

object = 3 of same color

$$r = \left\lceil \frac{N}{k} \right\rceil$$

N: total no of items

k: Boxes

r: number we want to guarantee in one box.

$\left\lceil \frac{N}{k} \right\rceil$

(this tells us the min no of items in the most filled (overstuffed box).

$$\left\lceil \frac{N}{2} \right\rceil = 3$$

$N = 5$

Permutations :: $^nP_r = P(n,r) = \frac{n!}{(n-r)!} = \frac{n(n-1)(n-2)(n-3)\dots(n-r)!}{(n-r)!}$

↳ method of arranging

r → permutation

if we have n distinct things (like n different ppl, numbers) and we want to arrange r of them, then the total no of possible arrangements

$P(n,r) = n \times (n-1)(n-2)\dots\dots$

* In permutation order matters

$n = 5$ and $r = 3$

$P(5,3) = 5 \times 4 \times 3 = 60$ there are 60 diff ways to arrange 3 ppl out of 5.

e.g $S = \{a, b, c\}$ $n = 3$

find all possible pairs (arrangements of 2 items) we can make from S.

$r = 2$ $3 \times 2 \rightarrow (6)$

ab →
ac
ba → order makes diff things
bc
ca
cb

$P(n,r) = \frac{n!}{(n-r)!}$ is same as $P(n,r) = n \times (n-1)(n-2)\dots\dots$

↳ total objects

↓ don't use this formula when arranging all n items -> Special condition like items must stay together

↓ this formula is used when we are arranging only r items out of n. (not all of them).

In permutation every arrangement must be unique.

e.g BALLOON

$n = 7$

L repeats = 2

O repeats = 2

total possible arrangements (ans) $\longrightarrow \frac{7!}{2! \times 2!} = 1260$

combinations → order doesn't matter. | we are choosing things |

$$C(n,r) = \frac{n!}{r!\,(n-r)!}$$

$$C(n,r) = C(n,n-r)$$

e.g ABC | CAB

$P = 2$

$C = 1$ ↗ order doesn't matter so ABC means the same as CAB.

Note: in combinations choosing r things is equal to skipping n-r things to skip.

You have 10 students and need to pick 4 only.

$$= \frac{10!}{4!\,(10-4)!} = 210$$

$$= \frac{10!}{6!\,(10-6)!} = 210$$

Q How many bit strings of length n contain exactly r 1's.

↓ (0 or 1) → C(n,r)

meaning only count 1 that can come in any order in a string.

$5x \equiv 3 \pmod 7$

$1 = 5 - 2 + 2$

$1 = 5 - 2(7 - 5 \times 1)$

$1 = 5 - 2 \times 7 + 5 \times 2$

$1 = 5 \times 2 - 2 \times 7$

$1 = 5 \times 2 + 5 - 2 \times 7$

$1 = 5 \times 3 - 2 \times 7$

$5 \times 3 \equiv 1 \pmod 7$

$gc \to 1$

$1 = 5 \times 1 + 2$

$2 = 7 - 5 \times 1$

V

$n \equiv 9 \pmod 7$

$n - 2$

$7 \sqrt{9}$ 1 7 2

Choosing → order doesn't matter

Arranging → order matters

! → done when order matters among all chosen items.

when numbers are involved order matters.

→ line, rank, password no → Permutation

→ committee, selection, group → combination

Bitstrings of length 8 exactly 3 ones and first bit is 1.

(1) _ _ _ _ _ _ _

7 positions out 2 1s

$= C(7,2)$

Ordered [selections]

$C(n,r) \times r!$

(Binomial coefficients and identities:-)

C++ → 28
Python → 26
Java → 32

14 → Both (Python + C++)
12 → Both (Java + Python)
10 → Both (C++ + Java)

$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$

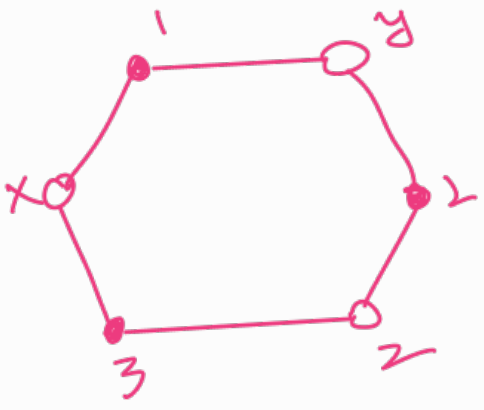
↳ Use this when sets overlap and the word "at least" is present

Recurrence Relation.

n (coefficient of n) + constant term = 0

$An + B = 2An - 2A + 2B + n + 5$

Bipartite graph → A graph made from two sets two sets. represent diff things where each consecutive vertex is part of a diff set and no two same consecutive vertices meet



A = {1, 2, 3}

B = {x, y, z}

general equation for Repeated roots

$a_n = (C_1 + C_2 n) r^n$ → Degree 2 $(r-3)^2 = 0$

$a_n = (C_1) r^n$ → Degree 1 $(r-3) = 0$

$a_n = (C_1 + C_2 n + C_3 n^2) r^n$

Characteristic eq

$(r-3)^3 = 0$

Detour to Ch: 2

nok: empty sets and the set itself are members of powerset.

contains all the subset members.

Set: $\{0,1,2\}$

Powerset: $\{\phi, 0, 1, 2, 01, 02, 12, 012\}$

Powerset of of ϕ - $\{\phi, \phi\}$

$\{1,3,5\} = \{3,5,7\} = \{1,3,3,5,5,5\}$

Function → takes in input and gives exactly one output

$f: A \to B$

↓ domain (input)

↓ output co-domain

Range → $\{$ min codomain - max codomain $\}$

One to one

→ Every distinct input → gives distinct output

Cardinality of a set: → no of elements in a set.

$A = \{1,2,3\} = 3$

Two sets have same cardinality if there is 1 to 1 correspondence.

$A = \{1,2,3\}$ $B = \{a,b,c\}$

$(1-a, 2-b, 3-c)$

Less than or equal cardinality: $|A| \leq |B|$

$A = \{x, y\}$ $B = \{1, 2, 3\}$

$x \to 1$ $y \to 2$

Strictly less cardinality $|A| < |B|$

meaning A has definitely fewer elements than B

$A: \{x, y\}$

$B: \{1, 2, 3\}$

$|A| < |B|$

countable sets: → finite $A = \{1, 2, 3, 4\}$

→ Infinite $N = \{1, 2, 3 \ldots\}$

Natural no $0, 1, 2 \ldots$

↓

But can be

Listed like $\langle 1, 2, 3 \ldots \rangle$

↘ Natural no

countably infinite means for each element we can map a natural no.

$\{0, 2, 4, 6, 8 \ldots\}$

↓ ↓ ↓ ↓ ↓

1 2 3 4 5

Set of all Integers

$\{-2, \ldots -3, -2, -1 \ldots 0, 1, 2, 3 \ldots \infty\}$

↓ ↓ ↓ ↓ ↓

5 3 1 2 4

Set of real nos.

$\{-\infty, -3, -2, -1, 0, 1, 2, 3 \cdots \infty\}$

contains $\{0.9, 0.10, 0.11$

NOT intuitively countable

cardinality of a countably infinite set

$|S| = \aleph_0$

Positive Rational numbers are countable

$\frac{p}{q} \quad p, q \in Z^+$

$1, \frac{1}{2}, 2, \frac{1}{3}, \frac{2}{3}$

$\frac{1}{1}$	$\frac{2}{1}$	$\frac{3}{1}$	$\frac{4}{1}$	$\frac{5}{1}$	$\cdots$
$\frac{1}{2}$	$\frac{2}{2}$	$\frac{3}{2}$	$\frac{4}{2}$	$\frac{5}{2}$	$\cdots$
$\frac{1}{3}$	$\frac{2}{3}$	$\frac{3}{3}$	$\frac{4}{3}$	$\frac{5}{3}$	$\cdots$
$\frac{1}{4}$	$\frac{2}{4}$	$\frac{3}{4}$	$\frac{4}{4}$	$\frac{5}{4}$	$\cdots$
$\frac{1}{5}$	$\frac{2}{5}$	$\frac{3}{5}$	$\frac{4}{5}$	$\frac{5}{5}$	$\cdots$

Avoid Repeated Rational numbers

Binomial Theorem → $\binom{n}{0}x^n y^0 + \binom{n}{1}x^{n-1}y^1 +$

$\binom{n}{2}x^{n-2}y^2 \cdots$

Expansion of $(x+y)^y$

$n = 4$

$\binom{4}{0}x^4y^0 + \binom{4}{1}x^3y + \binom{4}{2}xy^3 + \binom{4}{4}y^4$

$\downarrow$

$\frac{n!}{r!(n-r)!}$

general term

$\binom{n}{r}x^{n-r}y^r$